

From Cheap to Chic:

Enhancing Music Playback Quality of Budget Earphones via Hardware-Aware Learning

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INTRODUCTION

Low-end earphones are ubiquitous but deliver poor music quality due to hardware limitations.



46%

of earphones cost less than USD 50



566M

earphones shipped globally in 2024

EQ-based methods rely on frequency response curves (FRCs) measured under ideal conditions and fail to capture real-world non-linear distortion.

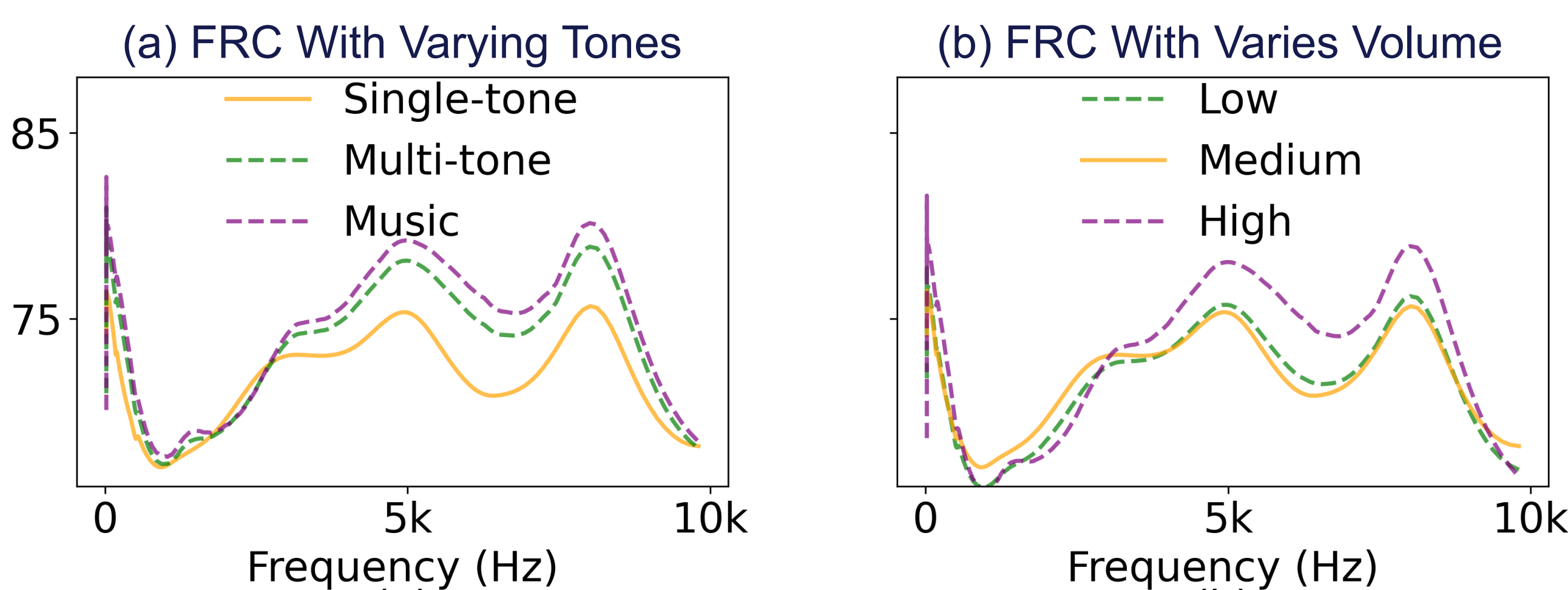


Our Goal:

Learn to compensate hardware-induced distortions in the digital domain, without any hardware changes.

1. THE CHALLENGE

FRC is static and linear, while real hardware distortion is dynamic and nonlinear.



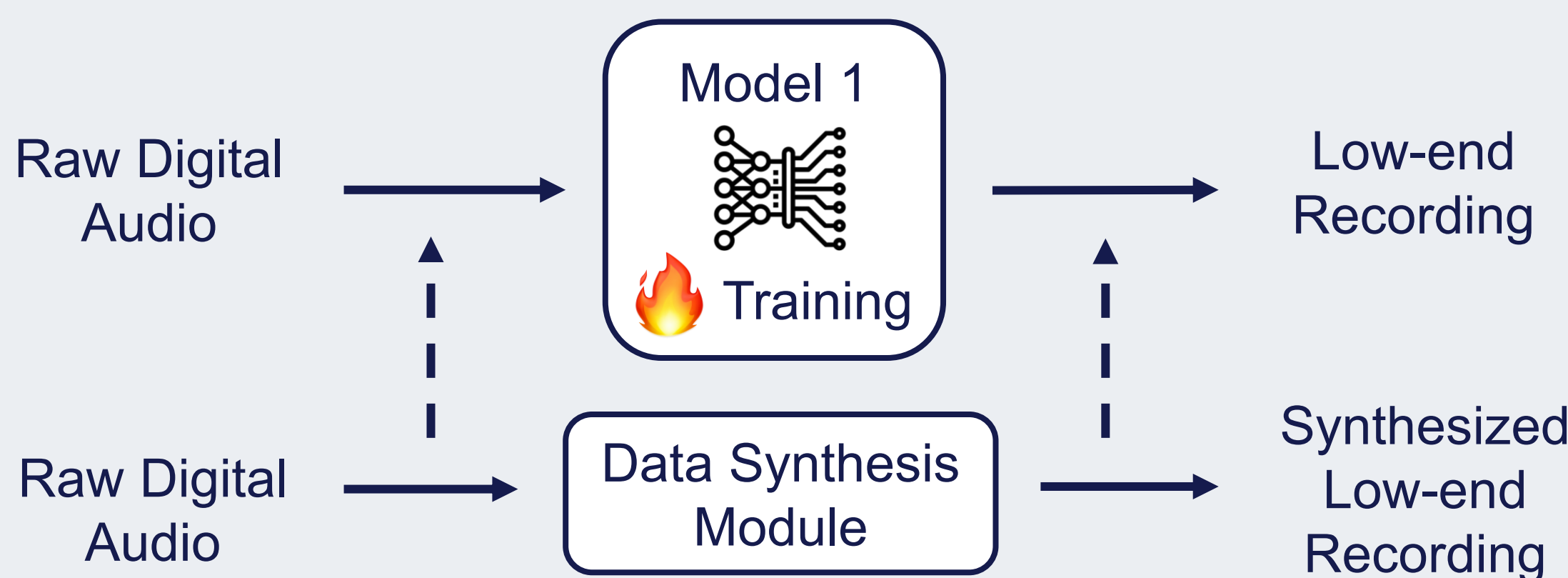
EQ Cannot Model:

- ✗ Harmonic Distortion
- ✗ Intermodulation Distortion
- ✗ Volume-Dependent Behaviors
- ✗ Complex Nonlinear Effects

2. CHEAP2CHIC: OUR APPROACH

A two-stage framework that learns speaker characteristics and enhances audio to compensate hardware limitations.

Stage 1: Low-end Speaker Characterization



Learn a neural proxy that simulates how a low-end earphone distorts audio.

Stage 2: Digital Audio Enhancement

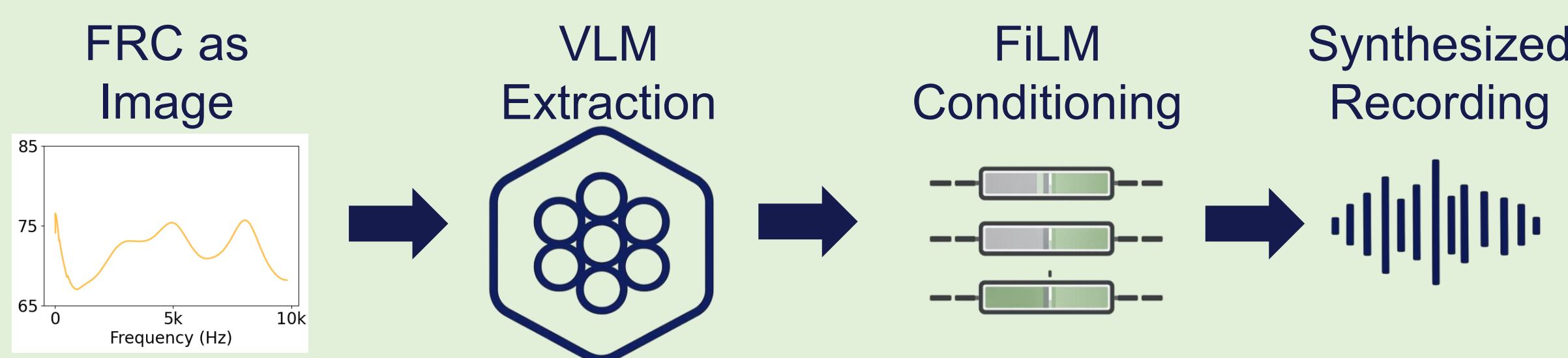


Learn to “pre-distort” the audio so that playback through the low-end device becomes high-quality.

Deployment: only Model 2 is required to generate enhanced digital audio.

Data Synthesis Module

Reduce data collection from 180 min to 30 min



- ✓ Treat the long sequence FRC as an image
- ✓ Vision-Language Model extracts perception-aware features
- ✓ FiLM conditions the generative model

3. EVALUATION & RESULTS

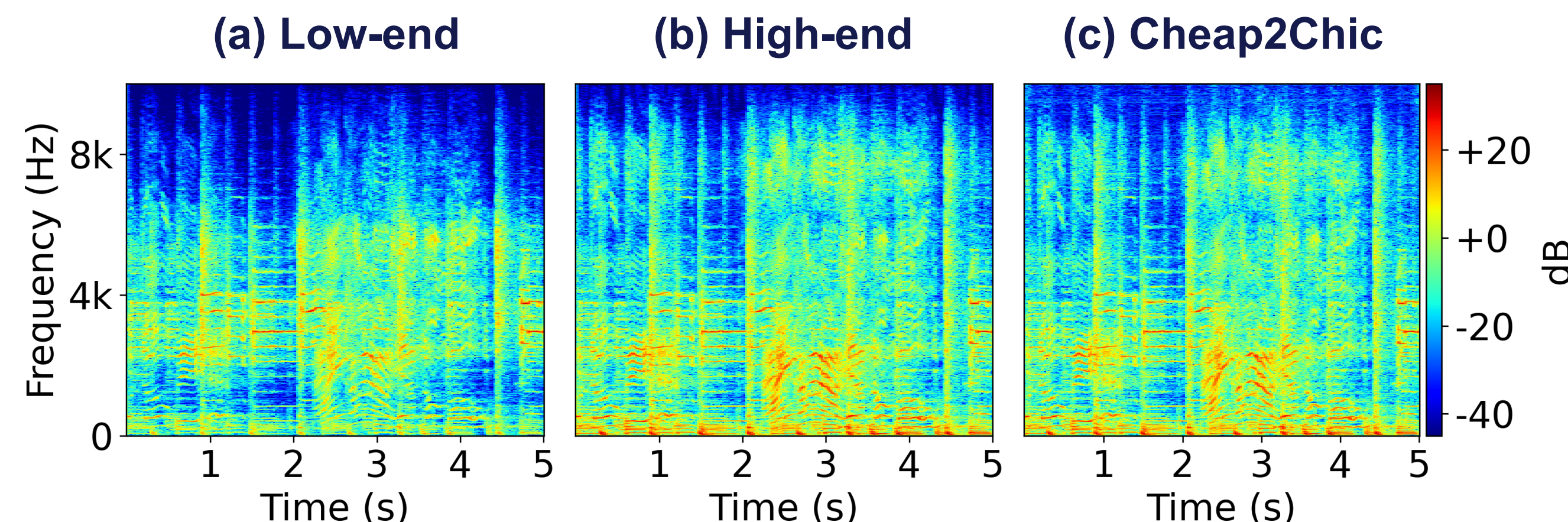
3.1 Objective Quality (relative differences with high-end)

System	SI-SDR↑	PEAQ↑	LSD↓	FAD↓
Low-end Playback	0.05	2.43	7.41	1.40
AutoEQ	9.75	3.03	6.60	0.78
DeFTAN-II	16.18	3.22	4.71	0.64
HS-TasNet	17.20	3.68	4.49	0.43
Limited-RealData	16.64	3.35	4.52	0.41
Full-RealData	20.74	4.04	2.66	0.11
Cheap2Chic (Ours)	20.56	3.91	2.93	0.14



Cheap2Chic Significantly outperforms EQ-based and deep baselines across all metrics (+20 dB SI-SDR over low-end).

3.2 Spectrogram Comparison



Cheap2Chic recovers spectral details and harmonic structure close to high-end.

3.3 Devices Generalization

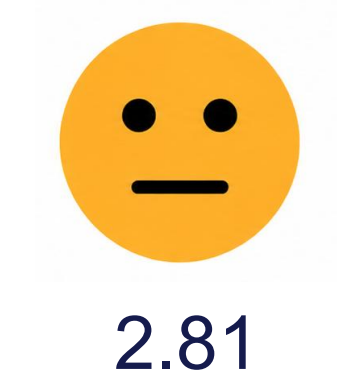
Intra-type Pairs				Cross-type Pairs			
In-ear	In-ear	Over-ear	Over-ear	In-ear	Over-ear	Over-ear	In-ear
+22.10 dB		+18.98 dB		+27.29 dB		+19.82 dB	

3.4 Subjective Listening Test (Mean Opinion Score)

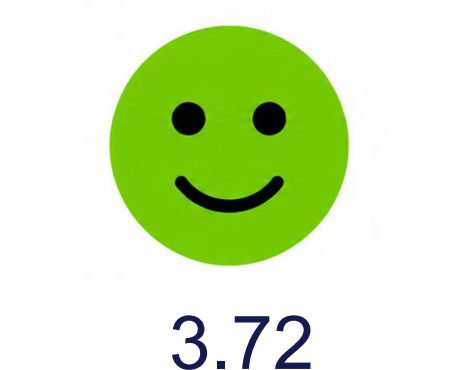


1 (Bad) – 5 (Excellent)

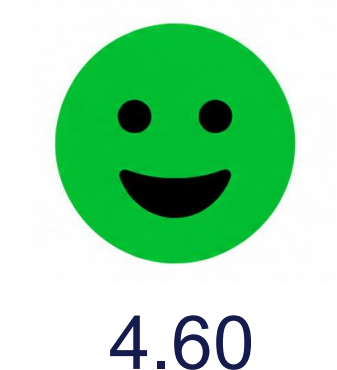
Low-end



Cheap2Chic



High-end



3.5 What Listeners Said



“Stronger bass and fuller sound.”



“Instruments are clearer and more natural.”



“Less harshness, more pleasant to listen.”

3.6 Latency and Power Consumption

Cheap2Chic runs on a Xiaomi 13 smartphone, processing each 5-second audio segment in 1.83 seconds with 6% extra battery consumption per hour.



Why Cheap2Chic?



Learns nonlinear Hardware behavior



Efficient & real-time on mobile



No hardware modification



Work across content & volume